# Joint SAGIN Resource Allocation and MAPF for Digital Twin Supported Remote Robotic Applications

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***Abstract*—Automation scenarios such as search and rescue, defense, and mining often operate in remote locations with unstable connectivity, making reliable communication a bottleneck for safe and efficient operations. Space–air–ground integrated networks (SAGIN) are a critical enabler, yet they introduce complex resource allocation challenges involving optical inter-satellite links and distributed computation for continuous digital twin (DT) verification. This study proposes a semantic X-haul framework for the co-design of DT resource distribution, optical backhaul, and aerial radio access networks. A multi-agent pathfinding algorithm suitable for complex environments is developed, using DTs for proactive control. We compare embodied AI versus cloud-based intelligence, evaluating trade-offs in unit cost, reliability, latency, and battery usage. Additionally, distributed and centralized DT architectures are analyzed in terms of delay, accuracy, and energy cost. Finally, by feeding DT predictions into the trajectory planning of a low-altitude aerial access point, we demonstrate that an intention-sharing method can improve overall energy expenditure. The results highlight the benefits of jointly designing computation and communications from algorithm design to resource allocation and exemplifies the unavoidable convergence between them.**

***Index Terms*—Satellite communications, robotics, cyber-physical systems, space-air-ground integrated networks, multi-agent path finding, 6G.**



Fig. 1: System model including SAGIN elements and cyber-physical systems operation in a remote location

## I. Introduction

- Digital Twins: The study in [1] proposes a method to utilize digital twins (DTs) for multi-domain resource management of SAGIN networks for ISAC purposes. This study also considers mapping errors of DTs due to imperfect sensing.
- SAGIN
- Ray-tracing
- Outage analysis: The studies in [2]–[4] provide outage analysis under a variety of system settings from CF-MIMO to FSO connectivity for SAGIN, where UAV is used either as a relay or access point. However, none of these articles take advantage of DTs. The study in [4] models satellite-UAV links as FSO and UAV-UE as RF links with MISO and Nakagami-m fading. This article provides analytical outage analysis under several practical constraints, such as outdated CSI and misalignment of FSO transcievers. The study in [2] investigates the affects of UAV position in the outage probability, which could be useful to analyze the affect of incorrect position estimation. The study in [3] provides uplink analysis and provides a method to optimize ground users' power transmit levels, which can be useful.

This work extends PANAMA's centralized training with decentralized execution framework [5] to distinguish communication delay from physical robot motion. During mission execution, delayed guidance and peer reports change the information available to a robot, while its local sensing and previously deployed policy remain available. During learning, delayed experience uploads restrict the learner's received data, and delayed model deployments cause robots to act with different versions of the shared policy. We formulate these effects through message timestamps, receiver caches, and model versions. The transport supplies the corresponding delays; their LEO-specific derivation is separate from the algorithm developed here.

## II. System Model

The system is as follows.

Let $\mathcal{N} = \{1, 2, \ldots, N\}$ denote the set of ground robots. Each robot $i \in \mathcal{N}$ generates a computationally intensive task modeled as a tuple $D_i = \{s_i, c_i, \tau_i\}$, where $s_i$ is the input data size in bits, $c_i$ is the required CPU cycles per bit, and $\tau_i$ is the maximum tolerable latency in seconds.

The system relies on the following decision variables:

- $x_i \in \{0, 1\}$: The offloading decision indicator. $x_i = 1$ if task $i$ is processed locally at the Drone-AP; $x_i = 0$ if the task is relayed to the Cloud.
- $P_i \geq 0$: The transmit power of robot $i$ on the Ground-to-Air (G2A) link.
- $f_{D,i} \geq 0$: The Drone CPU frequency allocated to task $i$.
- $f_{C,i} \geq 0$: The Cloud CPU frequency allocated to task $i$.

### A. Ground-to-Air (G2A) Channel

The G2A channel between the ground robots and the Drone-AP is modeled using a probabilistic Line-of-Sight (LoS) and Non-Line-of-Sight (NLoS) framework. The path loss for robot $i$, denoted $L_i^{\text{G2A}}$ in dB, is expressed as:

$$L_i^{\text{G2A}} = 20 \log_{10} \left( \frac{4\pi f_c d_i}{c} \right) + \eta \tag{1}$$

where $d_i$ is the 3D distance between robot $i$ and the drone, $f_c$ is the carrier frequency, $c$ is the speed of light, and $\eta$ is the additional attenuation factor depending on whether the link is LoS ($\eta_{\text{LoS}}$) or NLoS ($\eta_{\text{NLoS}}$).Add Al-Hourani's channel model here!!

The achievable uplink data rate is given by:

$$R_i^{\text{G2A}} = B_{\text{G2A}} \log_2 \left( 1 + \frac{P_i 10^{-L_i^{\text{G2A}}/10}}{N_0 B_{\text{G2A}}} \right) \tag{2}$$

where $B_{\text{G2A}}$ is the channel bandwidth and $N_0$ is the noise power spectral density.take the channel bw value from simulator!

### B. Air-to-Space (A2S) Channel

The backhaul link from the Drone-AP to the LEO satellite relies on a strong LoS channel. The path loss $L^{\text{A2S}}$ incorporates Free Space Path Loss and atmospheric attenuation $\xi_{\text{atm}}$:

$$L^{\text{A2S}} = 20 \log_{10} \left( \frac{4\pi f_{\text{sat}} d_{\text{sat}}^t}{c} \right) + \xi_{\text{atm}}, \tag{3}$$

where $d_{\text{sat}}^t$ indicates the distance between the satellite and the UAV-BS at time $t$.consider using $\bar{d}$ to indicate average d while uav and sat are associated!

The achievable A2S data rate for relayed tasks is:

$$R_i^{\text{A2S}} = B_{\text{A2S}} \log_2 \left( 1 + \frac{P_D |g|^2 10^{-L^{\text{A2S}}/10}}{N_0 B_{\text{A2S}}} \right) \tag{4}$$

where $P_D$ is the transmission power of the Drone-AP and $g$ models the Rician fading component.find the bw value and exact channel parameters!

We need to add space-air and air-to-ground downlink parameters. Only changing the tx power may be sufficient!

### C. Latency Model

The latency experienced by task $i$ depends on the offloading decision. If computed at the Drone-AP ($x_i = 1$):

$$T_{i,\text{Drone}} = \frac{s_i}{R_i^{\text{G2A}}} + \frac{s_i c_i}{f_{D,i}} \tag{5}$$

If the task is relayed to the Cloud ($x_i = 0$):

$$T_{i,\text{Cloud}} = \frac{s_i}{R_i^{\text{G2A}}} + \frac{s_i}{R_i^{\text{A2S}}} + T_{\text{prop}} + \frac{s_i c_i}{f_{C,i}} \tag{6}$$

where $T_{\text{prop}} \approx 2\bar{d}_{\text{sat}}/c$ represents the satellite round-trip propagation delay by using the average of the distance between the UAV and the satellite while associated. The total latency for task $i$ is thus:

$$T_i = x_i T_{i,\text{Drone}} + (1 - x_i) T_{i,\text{Cloud}} \tag{7}$$

### D. Energy Consumption Model

The transmission energy consumed by robot $i$ is:

$$E_i^{\text{tx}} = P_i \frac{s_i}{R_i^{\text{G2A}}} \tag{8}$$

confirm that this needs to be fixed for 3D placement The energy expended by the Drone-AP encompasses local computation and relaying transmissions. Let $\kappa_D$ denote the effective switched capacitance of the drone's computing architectureis this well known and widely used? how about GOP?. The drone energy cost for task $i$ is:

$$E_{\text{Drone},i} = x_i \kappa_D (f_{D,i})^2 s_i c_i + (1 - x_i) P_D \frac{s_i}{R_i^{\text{A2S}}} \tag{9}$$

### E. Timestamped Communication and Information Age

Let $n \in \mathbb{N}_0$ index joint robot actions, with decision time $t_n = n\Delta$ and step duration $\Delta > 0$. Robot $i$ occupies grid position $\mathbf{q}_i(n)$ and has a known goal $\mathbf{g}_i$. Its physical state is distinct from the information stored at a remote processor or another robot. Each message $m$ carries its source timestamp $g_m$, sequence or model version $\nu_m$, and complete payload. Its receiver obtains that payload at

$$b_m = g_m + \delta_m, \qquad \delta_m \geq 0, \tag{10}$$

where $\delta_m$ is the end-to-end delay supplied by the communication transport. A message with no completed delivery has $b_m = +\infty$. The algorithm does not prescribe a distribution for $\delta_m$ or a handover schedule. A subsequent LEO model determines these arrival times.

For a receiver stream $f$, let $\mathcal{M}_i^f(t_n)$ contain its messages fully received by $t_n$. We consider task guidance $f = G$, peer intentions $f = I_j$, and peer priorities $f = P_j$, for $j \in \mathcal{N}$. The accepted message and corresponding cache are

$$m_i^{f,*}(n) = \underset{m \in \mathcal{M}_i^f(t_n)}{\arg\max} \, (g_m, \nu_m), \tag{11}$$

$$\hat{z}_i^f(n) = z_{m_i^{f,*}(n)}, \tag{12}$$

where the maximum uses lexicographic timestamp–version order. An initial mission briefing initializes every cache at

$t_0$. If no newer message arrives, the stored value persists. An out-of-order older message cannot overwrite a newer accepted cache.

A relayed or aggregated payload also retains the timestamp of each source report, denoted $s_m^f$. Its information age is

$$A_i^f(n) = t_n - s^f_{m_i^{f,*}(n)}. \tag{13}$$

For guidance, this is the timestamp of the robot state used to compute the route. For intentions and priorities, it is the timestamp of the relevant peer report. A newly received message can therefore contain old source information. Aggregated peer age is the maximum of the included source ages.

Robots communicate through the UAV. For peer $j$ and receiver $i$, the transport delay $\delta^f_{j\to i}$ covers the report and returned update along the selected processing route. UAV processing permits local exchange without the satellite backhaul, whereas cloud processing requires backhaul forwarding. A cutoff prevents new deliveries on the affected route and increases cache age. It does not remove the last received peer intention or replace the robot's known goal. UAV positioning uses its own latest received robot-position reports, $\widehat{\mathbf{q}}_i^U(t)$, rather than instantaneous global positions unavailable to that receiver.

### F. Physical State and Available Decision Information

Self-position, locally sensed obstacles, and visible robot positions remain current at each action. Communication affects only the designated cached guidance, intentions, or priorities. The simulator resolves physical collisions, calculates task rewards, and checks mission completion using the true state. Cached priorities modify observation construction, not the physical transition rule. An age value is recorded for analysis; it is not an additional input to the retained policy. Thus the delay extension preserves the policy input dimensions while changing the timestamps of selected inputs.

## III. Optimization Problem Formulation

We formulate a Mixed-Integer Non-Linear Programming (MINLP) problem to jointly optimize the routing decisions, transmit powers, and computational resource allocations. The objective is to minimize a weighted sum of the robots' transmission energy, the system latency, and the drone's energy load. Let $\omega_1, \omega_2, \omega_3$ be user-defined weighting coefficients such that $\omega_1 + \omega_2 + \omega_3 = 1$.

The problem is mathematically formulated as:

$$\min_{\mathbf{x},\mathbf{P},\mathbf{f}_D,\mathbf{f}_C} \sum_{i=1}^{N} \left[\omega_1 E_i^{\text{tx}} + \omega_2 T_i + \omega_3 E_{\text{Drone},i}\right] \tag{14}$$

**Subject to:**

$$T_i \le \tau_i, \quad \forall i \in \mathcal{N} \tag{15}$$

$$0 \le P_i \le P_{\text{robot}}^{\max}, \quad \forall i \in \mathcal{N} \tag{16}$$

$$\sum_{i=1}^{N} x_i f_{D,i} \le F_{\text{Drone}}^{\max} \tag{17}$$

$$\sum_{i=1}^{N} (1 - x_i) f_{C,i} \le F_{\text{Cloud}}^{\max} \tag{18}$$

$$f_{D,i} \ge 0, \quad f_{C,i} \ge 0, \quad \forall i \in \mathcal{N} \tag{19}$$

$$x_i \in \{0, 1\}, \quad \forall i \in \mathcal{N} \tag{20}$$

Constraint (15) ensures that tasks are processed within their maximum tolerable latency limit. Constraint (16) bounds the ground robots' transmit power. Constraints (17) and (18) ensure that the total allocated CPU frequencies do not exceed the respective maximum capacities of the Drone-AP and the Cloud server. Finally, constraints (19) and (20) define the bounds and domains of the continuous and binary variables, respectively.

### A. PANAMA Policy with Cached Mission Information

We use PANAMA's shared dual-stream action-value network, Double DQN learner, prioritized experience replay (PER), task reward, and exploration mechanism [5]. Nominal learning parameters follow Table III of PANAMA, including discount $\gamma = 0.99$, AdamW learning rate $10^{-4}$, minibatch size $M = 64$, replay capacity $2\times10^6$, and target mixing coefficient $\eta_{\text{tar}} = 0.005$. The communication mechanism introduces no additional reward, imitation policy, or recovery action. We consider the task-policy channels for obstacles, visible robots, and communicated intentions, with UAV-mediated transport. Terrestrial radio maps and additional SNR-based shaping are outside this algorithmic extension.

An accepted guidance payload contains an absolute A* route $\mathcal{P}_i = (\mathbf{w}_{i,0}, \ldots, \mathbf{w}_{i,L})$ and reported distance $\widehat{d}_i$. The processor computes the route from the position in the generating report to $\mathbf{g}_i$, using the static obstacle map. While awaiting another guidance message, robot $i$ consumes waypoints it has already reached and retains the remaining absolute route. On late delivery, it also removes waypoints reached since that route's source timestamp. It neither recomputes A* locally nor translates old relative offsets as if they referred to its current position. The reported distance remains the received value until another guidance update arrives.

Let $\widehat{\mathbf{w}}_{i,k}(n)$ denote the $k$th remaining waypoint. The waypoint features use current self-position,

$$\widehat{\mathbf{u}}_{i,k}(n) = \text{clip}_{[-1,1]}\left(\frac{\widehat{\mathbf{w}}_{i,k}(n) - \mathbf{q}_i(n)}{h}\right), \tag{21}$$

where clipping is componentwise, $h = 7$ is the PANAMA field-of-view radius, and $k \in \{1, \ldots, K\}$ with $K = 5$. Unavailable waypoints contribute zero features. The retained task vector concatenates normalized reported distance, the local ever-reached-goal flag $f_i^g(n)$, and these waypoint pairs:

$$\widehat{\mathbf{v}}_i(n) = (\widehat{d}_i/d_i^0,\ f_i^g(n),\ \widehat{\mathbf{u}}_{i,1}, \ldots, \widehat{\mathbf{u}}_{i,K}), \tag{22}$$

with distance clipped to $[0, 1]$ and the initial briefing distance $d_i^0$ as normalizer. A nonpositive or nonfinite normalizer uses the map's larger dimension, as in the retained task implementation. The goal remains known even when the route or its reported distance is stale.

Let $\widehat{\mathcal{I}}_{j\to i}(n)$ be the last received absolute path segment of peer $j$, and let $\widehat{\rho}_{j\to i}(n)$ be its received priority, with smaller values indicating higher priority. Robot $i$ forms the intention channel from peers whose received priority precedes its own received priority, subject to PANAMA's communication range, local field of view, and received completion flag. A cached path cell $\mathbf{w}$ is placed relative to the current $\mathbf{q}_i(n)$. Writing $\mathcal{R}_i(n)$ for the eligible peers within PANAMA's communication range and $\widehat{f}^g_{j\to i}$ for their received completion flags, the contributing set is

$$\mathcal{H}_i(n) = \left\{ j \in \mathcal{R}_i(n) \setminus \{i\} : \begin{array}{l} \widehat{\rho}_{j\to i} < \widehat{\rho}_{i\to i}, \\ \widehat{f}^g_{j\to i} = 0 \end{array} \right\}. \tag{23}$$

For a relative field-of-view cell $\mathbf{z}$, the intention-channel value is

$$\widehat{O}^I_i(\mathbf{z}, n) = \mathbf{1}\left\{ \mathbf{q}_i(n) + \mathbf{z} \in \bigcup_{j\in\mathcal{H}_i(n)} \widehat{\mathcal{I}}_{j\to i}(n) \right\}. \tag{24}$$

Communication loss therefore leaves the peer's old segment in the observation, while local sensing can still reveal its actual nearby position. Denoting fresh obstacle and robot channels by $O^{\mathrm{obs}}_i(n)$ and $O^{\mathrm{rob}}_i(n)$, the policy input is

$$\widehat{o}_i(n) = \big([O^{\mathrm{obs}}_i, O^{\mathrm{rob}}_i, \widehat{O}^I_i], \widehat{\mathbf{v}}_i\big). \tag{25}$$

Each robot continues choosing among north, south, east, west, and wait using this available input. Communication unavailability does not itself mandate waiting or insert an external collision-recovery controller.

### B. Received Experiences and Continued Learner Operation

After joint action $n$, robot $i$ produces the complete transition

$$\zeta_i(n) = (\widehat{o}_i(n), a_i(n), r_i(n+1), \widehat{o}_i(n+1), z(n+1)), \tag{26}$$

where $z$ is the task simulator's terminal flag. The tuple is captured at $t_{n+1}$ together with robot, episode, joint-step, and deployed-version identifiers. Upload delay changes only its arrival time,

$$b^E_i(n) = t_{n+1} + \delta^E_i(n). \tag{27}$$

The transition's observations, action, reward, and terminal flag remain tied to their generating step, regardless of later robot motion or weight updates.

At learner time $t$, the received subset of the $N$ expected transitions from joint action $n$ is

$$\mathcal{U}_n(t) = \{i \in \mathcal{N} : b^E_i(n) \le t\}. \tag{28}$$

The replay buffer contains only fully received transitions, subject to its capacity and curriculum-phase retention rule. In particular,

$$\mathcal{B}(t) \subseteq \bigcup_{n:\, t_{n+1}\le t} \{\zeta_i(n) : i \in \mathcal{U}_n(t)\}. \tag{29}$$

The set on the right ranges over all generated joint actions. The learner does not require $\mathcal{U}_n(t) = \mathcal{N}$ before learning. It can use received experiences from unaffected robots and previously delivered replay samples while other uploads remain pending. If all current uploads are delayed, existing replay still supports updates. A learning opportunity is skipped only when insufficient replay data are available. Late experiences from a completed curriculum phase are excluded from the new phase's replay.

For each sampled transition $\zeta = (o, a, r, o', z)$, the retained Double DQN target and weighted Huber loss are

$$a^*_\zeta = \arg\max_{a'} Q_\theta(o', a'), \tag{30}$$

$$y_\zeta = r + \gamma(1 - z) Q_{\theta^-}(o', a^*_\zeta), \tag{31}$$

$$\mathcal{L}(\theta) = \frac{1}{M} \sum_{\zeta\in\mathcal{J}(t)} w_\zeta \, \ell_{\mathrm{H}}\big(Q_\theta(o, a) - y_\zeta\big), \tag{32}$$

where $\mathcal{J}(t)$ is a PER minibatch from $\mathcal{B}(t)$ and $w_\zeta$ is its importance-sampling weight. AdamW updates $\theta$ and the target follows $\theta^- \leftarrow (1 - \eta_{\mathrm{tar}})\theta^- + \eta_{\mathrm{tar}}\theta$. The delayed upload can change data availability and replay age, but does not replace the learning rule with an update based on incomplete tuples. The learner is off-policy: an experience collected with an older deployed version remains a complete training sample, and the learner evaluates its target with the current policy and target networks. Recording the generating version permits analysis of this mismatch without requiring every robot to hold the latest model before a learner update.

### C. Complete Model Deployment and Per-Robot Staleness

Let $\Theta^\ell$ be the immutable complete model snapshot published at time $u_\ell$, with increasing publication version $\ell$. For robot $i$, its completed downlink arrival is

$$b^W_i(\ell) = u_\ell + \delta^W_i(\ell). \tag{33}$$

Immediately before action $n$, robot $i$ uses

$$v_i(n) = \max\{\ell : b^W_i(\ell) \le t_n\}, \tag{34}$$

$$\widetilde{\theta}_i(n) = \Theta^{v_i(n)}, \tag{35}$$

with a complete initial model available at $t_0$. A receiver installs a newer snapshot atomically after full reception, before its next action. It never mixes layers, partially received parameters, or components of two versions. A late snapshot with $\ell \le v_i(n)$ cannot roll the deployed model back.

The action is selected by the standard epsilon-greedy rule,

$$a^*_i(n) = \arg\max_{a\in\mathcal{A}} Q_{\widetilde{\theta}_i(n)}(\widehat{o}_i(n), a), \tag{36}$$

$$\pi_i(a \mid \widehat{o}_i; n) = \frac{\epsilon_n}{|\mathcal{A}|} + (1 - \epsilon_n)\mathbf{1}\{a = a^*_i(n)\}. \tag{37}$$

Exploration retains PANAMA's schedule and depends on generated environment steps, not on delivered experience counts. A missing downlink update leaves $\widetilde{\theta}_i(n)$ unchanged while the robot continues acting. Other robots can install newer snapshots independently, and the centralized learner can continue optimizing its current model. Publications occur

Algorithm 1 PANAMA with timestamped information and learning transport

**Require:** Initial caches, model $\Theta^0$, replay $\mathcal{B}$, $\chi_L$

**for** each generated joint action $n$ **do**
  Process completed arrivals up to decision time $t_n$
  Accept newer information and complete model versions
  **for** each robot $i \in \mathcal{N}$ **do**
    Sense locally and consume reached cached waypoints
    Construct $\widehat{o}_i(n)$ and act with $\Theta^{v_i(n)}$
  **end for**
  Apply joint action using the true physical transition
  Advance event time and process completed arrivals
  **for** each robot $i \in \mathcal{N}$ **do**
    Submit timestamped mission reports to the transport
    Capture and queue the complete $\zeta_i(n)$
  **end for**
  Insert newly received eligible experiences into $\mathcal{B}$
  **if** $\chi_L = 1$ and $|\mathcal{B}| \geq M$ **then**
    Perform one standard PER/Double DQN update
  **end if**
  **if** $\chi_L = 1$ and $n+1$ is a multiple of $H$ **then**
    Publish and queue complete snapshot $\Theta^{\ell+1}$
  **end if**
**end for**

every $H$ generated joint actions, with the same $H$ in all learning conditions.

Let $\ell_{\mathrm{pub}}(t_n)$ be the latest published version. We separately measure model-version lag and model age,

$$L_i^W(n) = \ell_{\mathrm{pub}}(t_n) - v_i(n), \tag{38}$$

$$A_i^W(n) = t_n - u_{v_i(n)}. \tag{39}$$

Experience delivery age is $b_i^E(n) - t_{n+1}$; replay age at sampling time $t$ is $t - t_{n+1}$. These quantities distinguish old policy execution from old learning data without treating either as a halted mission.

### *D. Operation and Separation of Communication Effects*

Algorithm 1 summarizes the event-driven execution. The flag $\chi_L$ equals one when learning is enabled and zero during frozen-policy inference. One learning opportunity follows each generated joint action. Generated steps, delivered experiences, completed learner updates, and published models are counted independently. Delayed uploads cannot redefine the step counter or prevent other robots from contributing experiences. The simulator advances its communication clock with robot actions and processes arrivals at their timestamps, without a wall-clock sleep representing communication delay.

For frozen-policy mission analysis, all robots use one fixed snapshot and learning is disabled. The transport changes only the selected guidance, intention, or priority streams; unselected inputs remain current. For learning-transport analysis, mission information remains current and we compare immediate upload and deployment with delayed upload, delayed deployment, and their combination. The same initialization and nominal PANAMA parameters apply to those learning conditions. This separation attributes mission effects to stale communicated information and learning effects to received-data or deployed-model staleness. It makes no assumption that a communication delay necessarily reduces success.

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